

Name of the Experiment: Verification of Decoder using AND and NOT gate.

Objective: The objective of this experiment is to understand, realize and verify the logic gates.

Theory : A decoder is a combinational circuit. It has n inputs and a maximum $m = 2^n$ outputs. A decoder is identical to a demultiplexer without any data input. It performs operations which are exactly opposite to those of an encoder.

Block diagram:

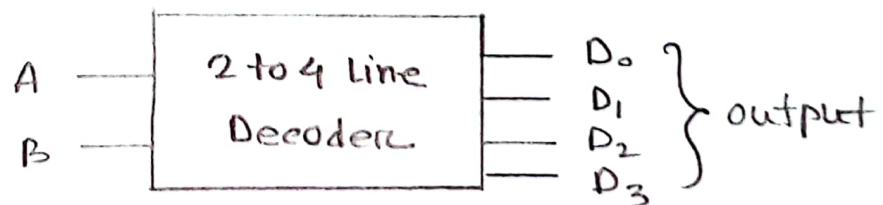


fig: Block diagram of 2 to 4 line decoder.

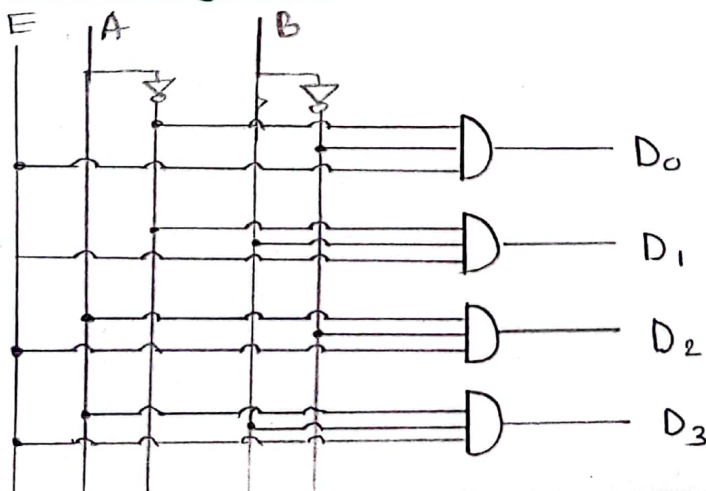
Apparatus :

- ① Bread-board
- ② Resistor - 4P
- ③ LED - 4P
- ④ Connecting wire
- ⑤ And AND gate (74LS08)
- ⑥ NOT gate (74LS04)

Truth table

Input		output			
A	B	D ₀	D ₁	D ₂	D ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Circuit diagram :



Procedure:

- Step 1: Construct the circuit of fig in the Bread-board. Every IC's pin '14' connected to +5V position of the DC power supply and pin 7 GND will be connected to the negative end of the power supply.
- Step 2: Then we use the pin configuration of the circuit diagram.
- Step 3: Changing the input (A,B).
- Step 4: Then we verify the result of different inputs.
- Step 5: Record what we observed when the input are change.

Result: If inputs are high then we connect the input directly to the AND gate. If inputs are low then we connected the inputs to the AND gate which we get from the output of Not gate.

Discussion: In this experiment we have checked all the gates of the IC's and found out the active gates. We also found out the position at which the IC's have proper supply.

We connected the V_{CC} and GND properly. finally we get output D_0, D_1, D_2, D_3 depending on input A, B .